Skills Summary

Proportional Relationships: Four Representations

Use this reference while completing your worksheet or when studying.

Graph to Equation

Rule: pick any clearly-marked point (x, y) on the line, compute $k = \frac{y}{x}$, and write $y = kx$. The line must pass through the origin, or the relationship is not proportional.

Example: A proportional graph passes through $(6, 9)$. Write its equation.
Step 1. Divide y by x at the marked point to get k, then write $y = kx$.
Step 2. $k = \frac{9}{6} = 1.5$.
$\Rightarrow$ $\boldsymbol{y = 1.5x}$

Try it: A proportional graph passes through $(4, 22)$. Write its equation.

check: $\boldsymbol{y = 5.5x}$

Table to Equation

Rule: compute $\frac{y}{x}$ for every row. If all rows agree, that shared ratio is k and the table's equation is $y = kx$. If the rows disagree, the table is not proportional and has no such equation.

Example: A table gives $(2, 4.5)$ and $(4, 9)$. Write its equation.
Step 1. Check each row's ratio — they must match before you may write $y = kx$.
Step 2. $\frac{4.5}{2} = 2.25$ and $\frac{9}{4} = 2.25$ — same ratio.
$\Rightarrow$ $\boldsymbol{y = 2.25x}$

Try it: A table gives $(3, 4.5)$ and $(6, 9)$. Write its equation.

check: $\boldsymbol{y = 1.5x}$

Equation to Context

Rule: in $y = kx$, the number k is the **unit rate** — what one unit of x is worth in the story (price per kg, km per hour). To answer a story question, substitute the given x and compute.

Example: $y = 2.5x$ gives the cost in dollars of x notebooks. Find the cost of 7 notebooks.
Step 1. The question hands you x — substitute it into the equation and multiply.
Step 2. $y = 2.5 \cdot 7 = 17.5$.
$\Rightarrow$ **17.5** dollars

Try it: $y = 4.5x$ gives kilometers walked in x hours. Find y when $x = 6$.

check: **27** km

Context to Graph

Rule: a steady rate in a story graphs as the line through the origin that rises by the rate for each unit step. Compute two or three points with $y = \text{rate} \times x$, plot them, and draw the line through $(0, 0)$.

Example: A cyclist rides 8 km each hour. Find the point on the graph at 3 hours.
Step 1. Multiply the rate by the time to get the height of the graph there.
Step 2. $y = 8 \cdot 3 = 24$, so the graph passes through the point $(3, 24)$.
$\Rightarrow$ **24** km at 3 hours

Try it: A kayaker paddles 7 km each hour. Find y on the graph at 4 hours.

check: **28** km